Inference in Discrete Graphical Models: Theory and Practice (full-day)

Abstract: Discrete energy minimization problems, in the form of factor graphs, Markov or Conditional Random Field models (MRF/CRF) are a mainstay of computer vision research. Their applications are diverse and range from image denoising, segmentation, motion estimation, and stereo, to object recognition and image editing.

The aim of this tutorial is to give researchers a clear guidance, which type of models are tractable and which optimization methods are best suited for these models. Besides theoretical aspects we will focus on practical questions, in particular on an optimization aware modeling.

1 Proposer (alphabetic order)

Thorsten Beier (thorsten.beier@iwr.uni-heidelberg.de)
Multidimensional Image Processing Group, Heidelberg University, Germany

received bachelor degree in physics from the University of Heidelberg in 2011, master degree in computer science from the University of Heidelberg in 2014. Since 2014 he is a PhD student at the University of Heidelberg. He is one of the main developers of OpenGM2, an open source library for inference in discrete graphical models. His main interests are optimization problems for multidimensional image segmentation and he is an author of recent papers dealing with large scale approximate optimization [4, 8].

Dr. Jörg Hendrik Kappes (kappes@math.uni-heidelberg.de)
Heidelberg Collaboratory for Image Processing, Heidelberg University, Germany

received diploma degree in computer science from the University of Mannheim in 2005 and it PhD degree from the Heidelberg University 2011. Since 2011 he is a PostDoc at Heidelberg University. His main research interests includes discrete optimization, computer vision and probabilistic graphical models. He regularly serves as reviewer for major computer vision and machine learning conferences (ICCV, CVPR, ECCV, ICML) and journals (TPAMI, IJCV) and is one of the major developers of OpenGM2, a free library for inference on discrete graphical models, and author of several well received papers dealing with inference on discrete graphical models including a recent benchmark paper on inference in discrete graphical models published in IJCV 2015. Google Scholar list 680 citations and a h-index of 14 for him.

Dr. Sebastian Nowozin (Sebastian.Nowozin@microsoft.com)
Microsoft Research Cambridge, UK

is researcher in the Machine Learning and Perception (MLP) group at Microsoft Research, Cambridge, UK. He received his PhD degree summa cum laude in 2009 from the Technical University of Berlin, Germany. His research is at the intersection of machine learning and computer vision. He regularly serves as area chair or program committee member at major conferences (NIPS, ICML, CVPR, ICCV, ECCV) and is reviewer for all major computer vision and machine learning journals (TPAMI, IJCV, JMLR, MLJ), as well as associate editor for IEEE TPAMI, IJCV, and JMLR. He has co-edited (with Suvrit Sra and Stephen Wright) a MIT Press volume on "Optimization for Machine Learning" and (with Peter Gehler, Jeremy Jancsary, and Christoph Lampert) a MIT Press volume on "Advanced Structured Prediction". He also published (with Christoph Lampert) a tutorial on structured prediction that has been presented at CVPR 2011 and CVPR 2012.

Prof. PhD. Carsten Rother (Carsten.Rother@tu-dresden.de)
Computer Vision Lab Dresden, Technische Universität Dresden, Germany

received the diploma degree in 1999 from the University of Karlsruhe, Germany, and the PhD degree in 2003 from the Royal Institute of Technology Stockholm, Sweden. From 2003 until 2013 he was researcher with Microsoft Research Cambridge, United Kingdom. Since October 2013 he is full professor at TU Dresden, heading the computer vision lab Dresden. One of his main research interests *is modelling, optimization and learning in structured models for various application tasks, including dense correspondence search, image segmentation, and BioImaging. In this field he has been teaching courses at university, at summer schools, and tutorials at conferences (CVPR 2010, ICCV 2009). His H-index is 51 and citation count over 13.000. He won awards at ACCV 2014, CVPR 2013, BMVC 2012, ACCV 2010, CHI 2007, CVPR 2005, and Indian Conference on Computer Vision 2010, and he was awarded the DAGM Olympus prize in 2009. He has co-edited (with Andrew Blake and Pushmeet Kohli) a MIT Press volume on "Markov Random Fields for Vision and Image Processing".*

Dr. Bogdan Savchynskyy (bogdan.savchynskyy@tu-dresden.de)
Computer Vision Lab Dresden, Technische Universität Dresden, Germany

graduated from the National Technical University of Ukraine "Kiev Polytechnic Institute" in 2002. In 2007 he defended his PhD Thesis at the National Academy of Sciences of Ukraine. In 2009-2014 he is a PostDoc at Heidelberg University, Germany, since 2015 - with Dresden University of Technology. Here he delivers a lecture course Machine Learning-2[1] *devoted to inference and learning for graphical models. His main interests are optimization problems for image and video analysis, in particular inference and learning with graphical models. He is a co-developer of OpenGM2, a free library for inference with discrete graphical models, co-author of the recent benchmark paper [6] and a number of works about exact and approximate inference for graphical models, e.g. [23, 19, 18, 22].*

The paper [23] of Bogdan Savchynskyy and Jörg Hendrik Kappes jointy authored with P. Swoboda and C. Schnörr recently received the **Best Student Paper Award** at CVPR'2014.

[1] `http://cvlab-dresden.de/courses/machine-learning-2-2015/`

1.1 Tutorial Outline

1. **Discrete Graphical Models** (50 min)
   a) Definitions and Notation
   b) Applications in Computer Vision
   c) Overview of Existing Software-packages

2. **Inference in Discrete Graphical Models : Theory and Practice** (150 min)
   a) Exact Inference Methods and Partial Optimality
   b) Inference Methods based on Relaxations
   c) Approximative and Move Making Methods
   d) Meta-Methods : Combining Methods to get a better overall performance.

3. **Learning in Discrete Graphical Models** (50 min)
   a) Problem Setting
   b) Maximum Likelihood based Methods
   c) Prediction-based Parameter Learning Methods

4. **Use Cases, Benchmarks and Useful Tricks** (50 min)
   a) Inside from benchmark studies (OpenGM [6] and MRF Middlebury [24])
   b) Common Use-cases and Textbook Models.
   c) How to deal with Huge Models and Higher-order Potentials?
   d) Anytime Behavior : Runtime vs. Accuracy
   e) Uncertainties in Solutions
   f) Models with Discrete and Continous Variables

5. **Conclusion** (20 min)

2 Tutorial Description

The proposed tutorial will focus on inference in discrete graphical models and its application in computer vision. The anticipated target audience are graduate students as well as researchers from fields where graphical models are used as a black box tool. We will give an overview of state-of-the-art methods together with compact but sufficient theoretical basis, which is required to understand advantages and restrictions of these methods. With this at hand, we will discuss the interplay between modeling and optimization and show how one can improve performance by using specialized methods instead of the most general ones.

In particular, we will cover *globally optimal methods* provided by general ILP-solvers, which recently have become significantly fast and popular. In this context we will discuss how partial optimality [12, 23, 22] can be obtained and used. Furthermore we will mention efficient exact solvers for important problem subclasses, such as max-flow for submodular energies or dynamic programming and perfect matching for acyclic and planar graphs respectively.

For *linear programming relaxations* we will give the fundamental theory and show relations between the corresponding dedicated solvers. This includes sub-gradient, smoothing and message passing based techniques. In this context we will also discuss tightening of the underling relaxation and obtaining sound stopping conditions for iterative inference algorithms.

The part of the tutorial entitled *Approximative and Move Making Methods* will be devoted to such methods as α-expansion, $\alpha\beta$-swap as well as fusion-moves and (block-) ICM.

We will conclude the inference part of the tutorial by showing how different methods can be combined into *meta-methods*, which often have the best overall performance.

An important part of this tutorial are common use-cases, problems and pitfalls, which will be covered by a separate lecture block with several live demos.

We then will turn to the problem of learning parameters of discrete graphical models. After giving the problem description we will introduce some classical learning methods and discuss problems that have to be considered in practice.

Due to the wide scope of the topic we would prefer to give a full day lecture as it was common to similar tutorials in the last years. We expect between 50 and 100 attendees.

We would prefer to give the tutorial either at the 12th or 17th of December.

Most relevant publications of the proposers are:

- Textbooks: [15]

- Benchmarks/Software: [24, 1, 7, 6]
- Global Optimal Solvers: [23, 19, 12, 11, 22]
- Linear Programming Relaxations:[17, 3, 21, 9, 18, 10, 10, 20]
- Move Making Methods: [13, 2, 4, 8, 2]
- Learning: [14, 5]

Course materials :

- All slides will be made available online
- Code of the discussed methods and shown examples is/will be available online

3 Related Tutorials

Due to the importants of graphical models nearly each of the mayor computer vision convergences had a tutorial on graphical models in the last years.

Higher Order Model Tutorial (ECCV 2014 - full day)
The main focus of this workshop were higher order models in connection with mean-field and graph-cut methods.
→ *We will cover a wider spectrum of methods*

Learning and inference in discrete graphical models (CVPR 2014 - half day)
This tutorial had covered inference and learning on graphical models. While the content was already dense for graduate students it was restricted to 'older' state-of-the-art inference methods (e.g. LBP, QPBO, GC, ICM, a-exp, ab-swap, Dual decomposition) only.
→ *We will spent more time on explaining fundamental concepts of latest inference methods and their impact in computer vision.*

Beyond MAP: Making Multiple Predictions: Diversity, DPPs and more. (CVPR 2013 - full day)
This full day tutorial discuss the problem of finding the M-Best solutions of a discrete graphical model.
→ *Our topic is a way more general. 'M-best' is a special subclass*

Graph Cut Based Optimization for Computer Vision (CVPR 2012 - half day)
This tutorial was only focused on graph cut based methods
→ *Our topic is more general. We will cover graph cut based methods as well, but not as deep as in this tutorial*

Structured Prediction and Learning in Computer Vision (CVPR 2011/2012 - full day)
This full day tutorial based on [16] had covered inference and learning on graphical models. Inference methods had been restricted to graph cut-based methods, ICM, α-Expansion and $\alpha\beta$-swap, message passing, simulated annealing, linear programming relaxations and branch and bound techniques.
→ *This tutorial is the closest one to our proposal. However we will focus on how to select proper inference methods in applications and more on inference than on learning. We also will include advances of the last 3 years.*

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